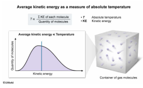
The absolute temperature of ideal gas molecules stored in a container is directly proportional to the:

- A. quantity of gas molecules. [7%]
- B. intermolecular forces between gas molecules. [4%]
- ✔ C. average kinetic energy of gas molecules. [82%]
- D. maximum velocity of gas molecules. [5%]

Correct
83% answered correctly

Explanation:



In common terms, **temperature (*T*)** is an objective measure of the relative "heat" or "cold" contained with an object or system. Temperature influences important physical properties like the physical state and reactivity of matter.

The **absolute temperature scale** is used to quantify the temperature of objects relative to **absolute zero**, the temperature at which matter achieves the lowest possible quantity of total energy (kinetic energy + potential energy). Absolute zero is considered "absolute" because it is the same for all matter, regardless of atomic structure or molecular configuration.

The unit of absolute temperature is the Kelvin (K), and the gradations of the absolute temperature scale are equal in size to those of the centigrade scale. The scales differ only in the placement of the zero value. Consequently, the temperature in Kelvin (T_K) relates to the temperature in Celsius (T_C) through the following expression:

$$T_K = T_C + 273.15$$

Fundamentally, kinetic energy is the key determinant of absolute temperature. Because temperature is typically sampled from multiple atoms or molecules, however, the temperature of matter reflects the average kinetic energy (KE) of a collection of atoms or molecules:

$$T \propto \frac{\sum \text{KE of each molecule}}{\text{Quantity of molecules}}$$

Therefore, the temperature of gas molecules moving throughout an enclosed container is determined not by the kinetic energy of any one gas molecule but rather by the average kinetic energy of all gas molecules.

(Choice A) The quantity of gas molecules is directly proportional to the pressure and volume, but not the temperature, of a gas.

(Choice B) Although significant intermolecular forces between gas molecules would influence the kinetic energy of those molecules, intermolecular forces are not observed between ideal gas molecules.

(Choice D) Gases exhibit a range of possible kinetic energy values at all temperatures. However, temperature is not related to the kinetic energy of individual molecules at extreme kinetic energy values, but rather the average kinetic energy of all molecules.

Educational objective:
The absolute temperature scale quantifies temperature relative to absolute zero, the lowest possible total energy of matter. The absolute temperature of any system is directly proportional to the average kinetic energy of molecules within the system.

Time spent: 0 sec
Subject: Physics

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System: Fluids